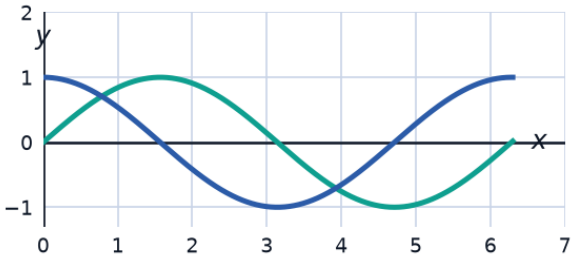
Further Integration Techniques

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

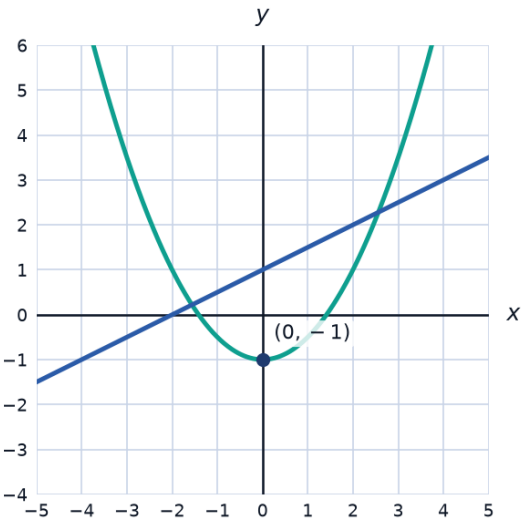
1 [Foundational] • Target: Finding the area between two curves using integration

Find the area of the region enclosed between $y = \sin x$ and $y = \cos x$ for $\pi/4 \le x \le 5\pi/4$ (the interval on which $\sin x \ge \cos x$).



2 [Foundational] • Target: Integration by substitution

Evaluate $\int 2x(x^2 + 1)^4\, dx$ using the substitution $u = x^2 + 1$.



3 [Foundational] • Target: Integration by substitution

Evaluate $\int_0^{\pi/2} \sin^3 x \cos x\, dx$ using the substitution $u = \sin x$.

4 [Foundational] • Target: Integration by parts

Evaluate $\int xe^x\, dx$ using integration by parts.

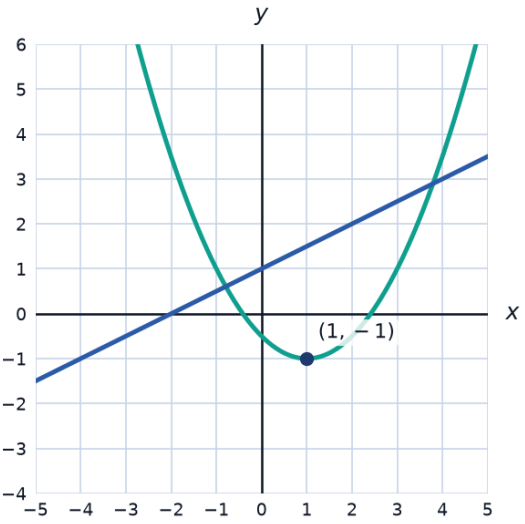
5 [Foundational] • Target: Integration by parts

Evaluate $\int x \ln x\, dx$ using integration by parts.

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6 [Foundational] • Target: Repeated integration by parts

Evaluate $\int x^2 e^x \, dx$, applying integration by parts twice.



7 [Foundational] • Target: Integration using trigonometric identities

Evaluate $\int \cos^2 x \, dx$ using the identity $\cos^2 x = (1 + \cos 2x)/2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Integration using partial fractions

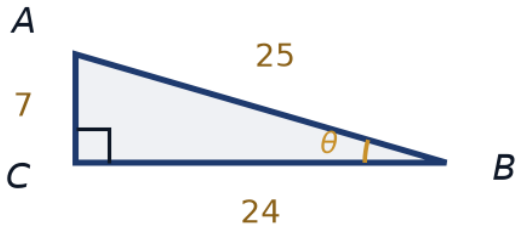
Evaluate $\int 1/(x^2 - 1) \, dx$ by first expressing the integrand in partial fractions.

9 [Intermediate] • Target: Evaluating a definite integral

Evaluate $\int_1^e \ln x / x \, dx$, using the substitution $u = \ln x$.

10 [Intermediate] • Target: Evaluating a definite integral

Evaluate $\int_0^{\pi} x \sin x \, dx$ using integration by parts.



11 [Intermediate] • Target: Integration involving inverse trigonometric results

Evaluate $\int 1/(1 + x^2) \, dx$, and use it to evaluate $\int_0^1 1/(1 + x^2) \, dx$.

12 [Intermediate] • Target: Integration by substitution with trigonometric functions

Evaluate $\int \sin^3 x \, dx$, using the identity $\sin^2 x = 1 - \cos^2 x$ and the substitution $u = \cos x$.

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13 [Intermediate] • Target: Integration by substitution with trigonometric functions

Evaluate $\int \tan x \, dx$, using the substitution $u = \cos x$.

14 [Intermediate] • Target: Integration by parts with a logarithmic integrand

Evaluate $\int_1^e \ln x \, dx$ using integration by parts.

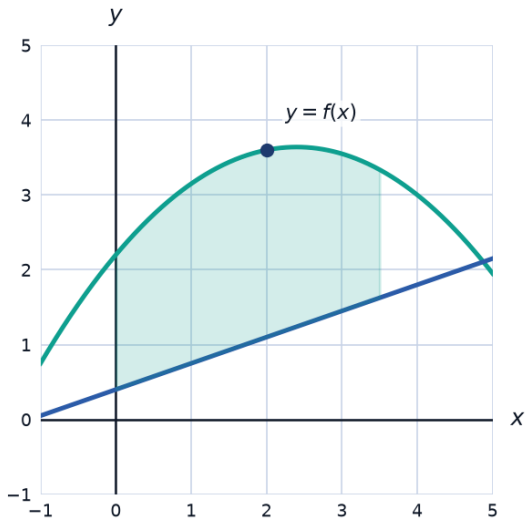
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Integration using a Weierstrass-style substitution • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Evaluate $\int 1/(5 + 4\cos x) \, dx$ using the substitution $t = \tan(x/2)$, given that this substitution transforms $\cos x = (1 - t^2)/(1 + t^2)$ and $dx = 2/(1 + t^2)dt$, leading to the simplified integrand $2/(9 + t^2)$.

16 [Challenging] • Target: Integration by parts with a trigonometric-exponential product • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Evaluate $\int e^x \cos x \, dx$, applying integration by parts twice and solving the resulting equation for the original integral.



17 [Challenging] • Target: Finding the area between two curves using integration • Cross-Lesson: 4.02 The Poisson Distribution

Evaluate $\int x^2 \ln x \, dx$ using integration by parts.

18 [Challenging] • Target: Finding the area between two curves using integration • Cross-Lesson: 5.02 Differential Equations

A particle's velocity is given by $v(t) = t\cos t$ metres per second. Find the particle's displacement over the interval $0 \le t \le \pi$, using integration by parts.

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19 [Challenging] • Target: Finding the area between two curves using integration • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

This question has three parts. (a) Use the substitution $u = x^2 + 4$ to find $\int x/(x^2+4)\,dx$. (b) Use integration by parts to find $\int \arctan x\,dx$. (c) Hence, using the result of part (b) and the standard result $\int 1/(1+x^2)dx = \arctan x + C$, evaluate $\int_0^1 \arctan x\,dx$, giving an exact answer in terms of π and $\ln$.

20 [Challenging] • Target: Reduction formulas • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Given the reduction formula $I_n = \int x^n e^x\,dx = x^n e^x - nI_{n-1}$ (obtained via integration by parts), and that $I_0 = e^x + C$, use it to find $I_2 = \int x^2 e^x\,dx$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Choosing an appropriate integration technique

Which integration technique is most appropriate for evaluating $\int x^3 \sin x\,dx$?

a. Integration by parts, applied three times (reducing the power of x each time)

b. Substitution $u = \sin x$

c. Partial fractions

d. The double-angle identity for $\sin$

22 [Foundational] • Target: Integration by Parts (Repeated & Cyclic)

Evaluate $\int x^2 \cos x dx$ using integration by parts.

a. $x^2 \sin x + 2x\cos x - 2\sin x + C$

b. $x^2 \sin x - 2x\cos x + 2\sin x + C$

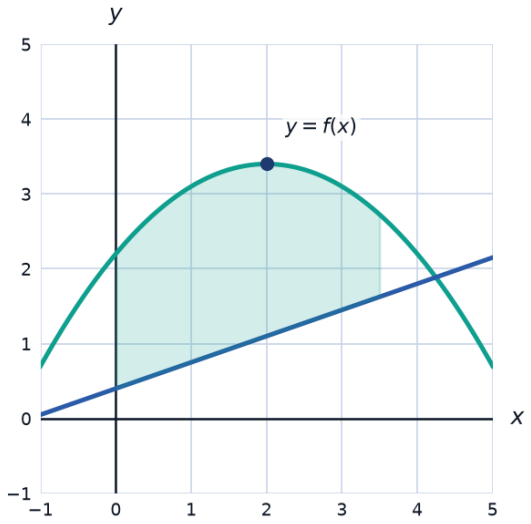
c. $-x^2 \sin x + 2x\cos x + C$

d. $\frac{1}{3}x^3 \sin x + C$

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23 [Foundational] • Target: Integration by Parts (Repeated & Cyclic)

Evaluate the cyclic integral $I = \int e^x \sin x \, dx$.



a. $\frac{1}{2}e^x(\sin x - \cos x) + C$

b. $\frac{1}{2}e^x(\sin x + \cos x) + C$

c. $e^x(\sin x - \cos x) + C$

d. $-e^x \cos x + C$

24 [Foundational] • Target: Integration by Parts (Repeated & Cyclic)

Evaluate $\int_1^e \ln x \, dx$.

a. 1

b. $e - 1$

c. e

d. 0

25 [Foundational] • Target: Trigonometric Substitutions

Which trigonometric substitution is best suited to evaluate $\int \frac{1}{\sqrt{16 - x^2}} dx$?

a. $x = 4\sin\theta$

b. $x = 4\tan\theta$

c. $x = 4\sec\theta$

d. $x = 16\sin\theta$

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26 [Foundational] • Target: Trigonometric Substitutions

Evaluate $\int \frac{1}{x^2+16} dx$.

a. $\frac{1}{4}\arctan\left(\frac{x}{4}\right)+C$

b. $\arctan\left(\frac{x}{4}\right)+C$

c. $\frac{1}{16}\arctan\left(\frac{x}{16}\right)+C$

d. $\frac{1}{4}\ln|x^2+16|+C$

27 [Foundational] • Target: Trigonometric Substitutions

Which substitution evaluates $\int \sqrt{x^2-9}\,dx$?

a. $x=3\sec\theta$

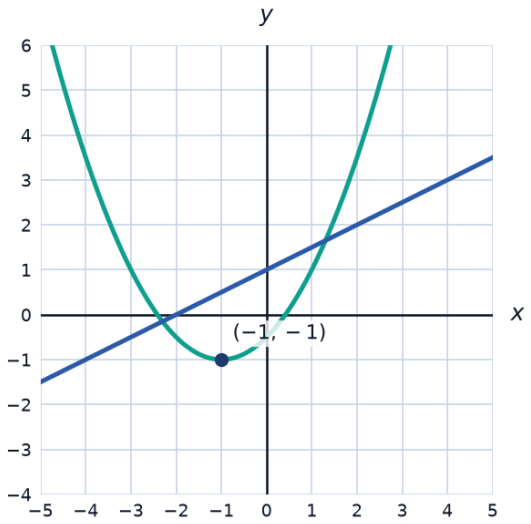
b. $x=3\sin\theta$

c. $x=3\tan\theta$

d. $x=9\sec\theta$

28 [Foundational] • Target: Integrals of Powers of Trigonometric Functions

Evaluate $\int \sin^3 x dx$.



a. $-\cos x+\frac{1}{3}\cos^3 x+C$

b. $\cos x-\frac{1}{3}\cos^3 x+C$

c. $\frac{1}{4}\sin^4 x+C$

d. $-\frac{1}{3}\cos^3 x+C$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Integrals of Powers of Trigonometric Functions**

Evaluate $\int \cos^2 x dx$.

a. $\frac{1}{2}x + \frac{1}{4}\sin(2x) + C$

b. $\frac{1}{2}x - \frac{1}{4}\sin(2x) + C$

c. $\frac{1}{3}\cos^3 x + C$

d. $\frac{1}{2}x + \sin(2x) + C$

30 **[Intermediate]** • **Target: Integrals of Powers of Trigonometric Functions**

Evaluate $\int \tan^2 x dx$.

a. $\tan x - x + C$

b. $\sec^2 x + C$

c. $\tan x + x + C$

d. $\frac{1}{3}\tan^3 x + C$

31 **[Intermediate]** • **Target: Integration by Substitution (Advanced Forms)**

Evaluate $\int \frac{x}{\sqrt{x+1}} dx$ using substitution $u = x + 1$.

a. $\frac{2}{3}(x+1)^{3/2} - 2(x+1)^{1/2} + C$

b. $\frac{2}{3}(x+1)^{3/2} + 2(x+1)^{1/2} + C$

c. $(x+1)^{3/2} - (x+1)^{1/2} + C$

d. $\frac{1}{2}(x+1)^2 + C$

32 **[Intermediate]** • **Target: Integration by Substitution (Advanced Forms)**

Evaluate $\int_0^1 \frac{e^x}{1+e^x} dx$.

a. $\ln\left(\frac{1+e}{2}\right)$

b. $\ln(1+e)$

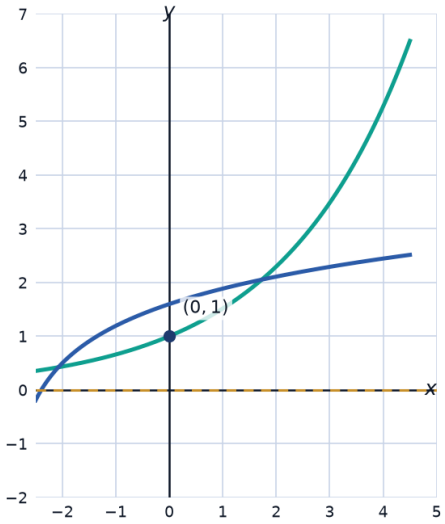
c. $e - 1$

d. $\frac{e}{2}$

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33 **[Intermediate]** • **Target: Integration by Substitution (Advanced Forms)**

Evaluate $\int \frac{1}{x\ln x}dx$ for $x > 1$.



a. $\ln|\ln x| + C$

b. $\frac{1}{2}(\ln x)^2 + C$

c. $\frac{1}{\ln x} + C$

d. $x\ln x - x + C$

34 **[Intermediate]** • **Target: Integration by substitution with a linear inner function**

Consider the function $f(x) = \frac{\sin(2x)}{\cos(x)}$. For the domain $x \in [0, 2\pi]$ such that $\cos(x) eq 0$, find the exact value of the definite integral $\int_0^{\pi/2} f(x)dx$.

a. $\ln(2)$

b. 1

c. 2

d. $\ln(1)$

35 **[Intermediate]** • **Target: Integration using the sine double-angle identity**

If $f(x) = \frac{7}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).

a. $\sqrt{7}$

b. $\sqrt{10}$

c. $\sqrt{5}$

d. $\sqrt{6}$

36 **[Intermediate]** • **Target: Volume of revolution using integration**

If $f(x) = \frac{8}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).

a. 3

b. $\sqrt{8}$

c. $\sqrt{11}$

d. $\sqrt{6}$

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37 [Intermediate] • Target: Substitution with a shifting definite integral bound

If $f(x) = \frac{5}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).

a. $\sqrt{8}$

b. 2

c. $\sqrt{5}$

d. $\sqrt{7}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Choosing an appropriate integration technique • Cross-Lesson: 5.02 Differential Equations

If $f(x) = \frac{3}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).

a. $\sqrt{3}$

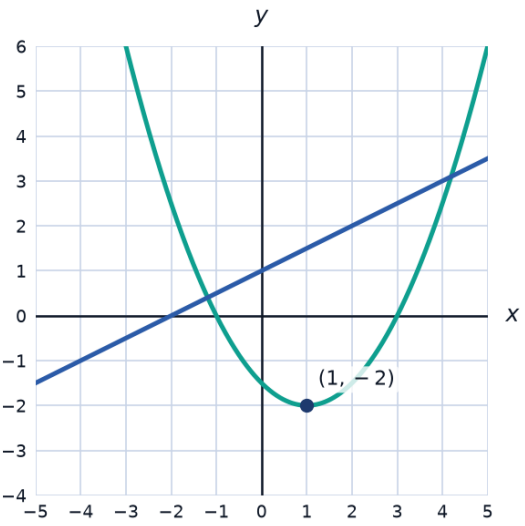
b. $\sqrt{5}$

c. 2

d. $\sqrt{6}$

39 [Challenging] • Target: Integration by Parts (Repeated & Cyclic) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

If $f(x) = \frac{4}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).



a. 2

b. 1

c. 3

d. 4

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40 [Challenging] • Target: Trigonometric Substitutions • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Integration by parts chooses u such that:

a. u is simple to integrate

b. du/dx is simpler than u

c. dv is the whole integrand

d. u contains e^x always

41 [Challenging] • Target: Integrals of Powers of Trigonometric Functions • Cross-Lesson: 6.01 Matrices (AI HL)

Find the angle between lines with direction vectors $\mathbf{d_1} = (1; 4; 8)$ and $\mathbf{d_2} = (4; -1; 0)$.

a. 90°

b. 45°

c. 60°

d. 30°

42 [Challenging] • Target: Integration by Substitution (Advanced Forms) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Find the angle between lines with direction vectors $\mathbf{d_1} = (4; 4; 7)$ and $\mathbf{d_2} = (4; -4; 0)$.

a. 45°

b. 30°

c. 60°

d. 90°

43 [Challenging] • Target: Integration by substitution • Cross-Lesson: 6.03 Markov Chains (AI HL)

Find the angle between lines with direction vectors $\mathbf{d_1} = (1; 2; 2)$ and $\mathbf{d_2} = (2; -1; 0)$.

a. 60°

b. 90°

c. 45°

d. 30°

44 [Challenging] • Target: Integration by parts • Cross-Lesson: 1.01 Proof by Mathematical Induction

Find the angle between lines with direction vectors $\mathbf{d_1} = (3; 4; 12)$ and $\mathbf{d_2} = (4; -3; 0)$.

a. 90°

b. 60°

c. 30°

d. 45°

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45 [Challenging] • Target: Repeated integration by parts • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Find the angle between lines with direction vectors $\mathbf{d_1} = (6; 2; 3)$ and $\mathbf{d_2} = (2; -6; 0)$.

a. 90°

b. 45°

c. 60°

d. 30°

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